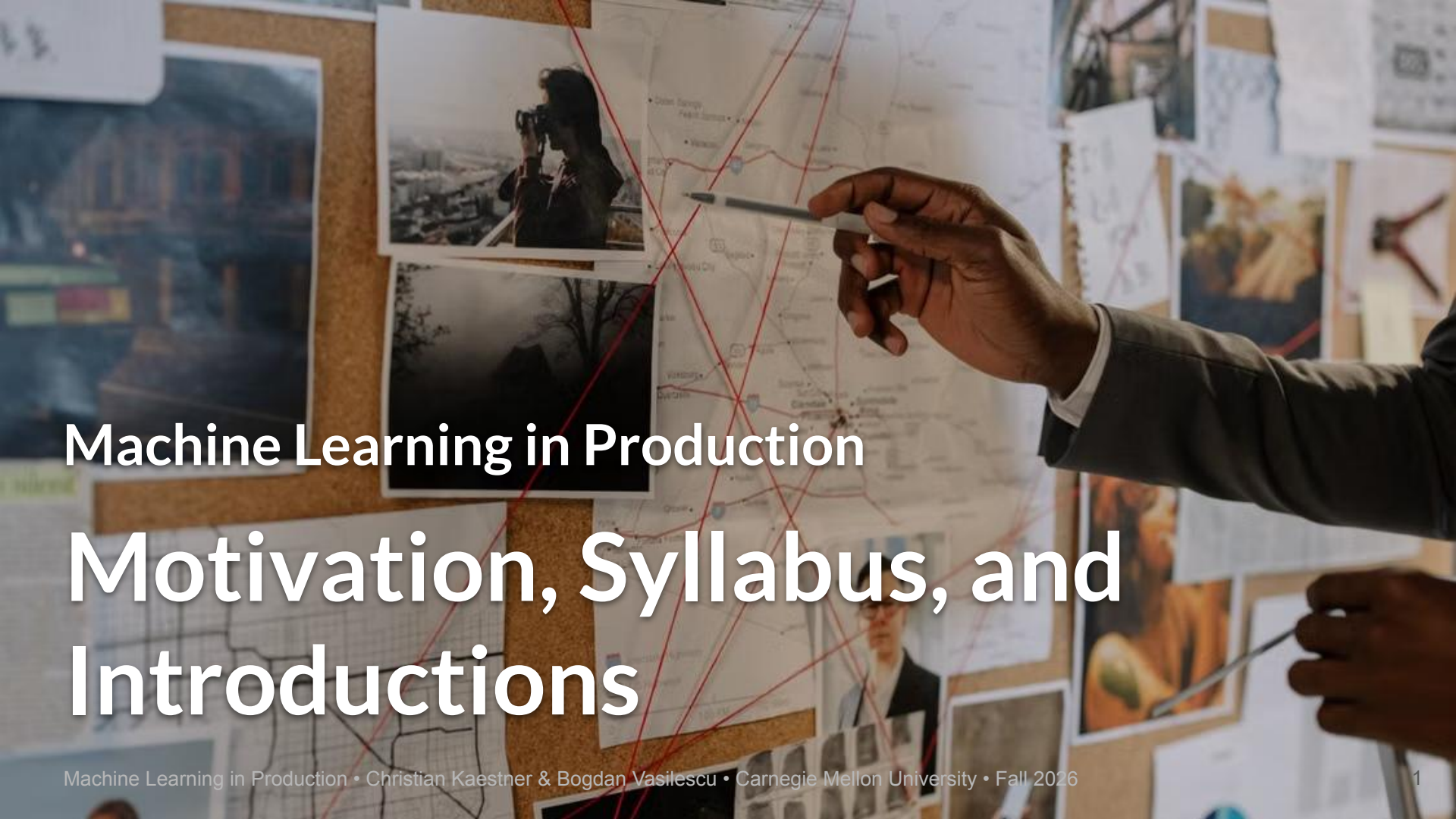


A hand in a grey suit sleeve points with a pen at a map on a corkboard. The corkboard is covered with various items: a photo of a person taking a picture, a dark landscape photo, a map of the United States with red string connecting different cities, a photo of a person in a suit, and a photo of a person in a yellow shirt. The text 'Machine Learning in Production Motivation, Syllabus, and Introductions' is overlaid on the image.

Machine Learning in Production Motivation, Syllabus, and Introductions

Slack

- We use Slack for this course, including during lectures
- See signup link on Canvas (you can do this right now!)
- Setup the ability to read/post to Slack during lecture.

Introductions

Before the next lecture, introduce yourself in Slack channel #social:

- Your (preferred) name
- In 1-2 sentences, describe your data science background and goals (e.g., coursework, internships, work experience)
- In 1-2 sentences, describe your software engineering background, if any, and goals (e.g., coursework, internships, work experience)
- The kind of job you are aiming for after graduation
- A hobby or a favorite activity outside school

Learning Goals

- Understand how ML components are parts of larger systems
- Illustrate the challenges in engineering an ML-enabled system beyond accuracy
- Explain the role of specifications and their lack in machine learning and the relationship to deductive and inductive reasoning
- Summarize the respective goals and challenges of software engineers vs data scientists
- Explain the concept and relevance of “T-shaped people”
- Understand the basic mechanics of this course.

Case Study: Detecting AI-Generated Text

Context: Machine-Generated Text Detection Research

Mitchell et al. "DetectGPT: Zero-shot machine-generated text detection using probability curvature." ICML 2023.

The increasing fluency and widespread usage of large language models (LLMs) highlight the desirability of corresponding tools aiding detection of LLM-generated text. In this paper, we identify a property of the structure of an LLM's probability function that is useful for such detection. Specifically, we demonstrate that text sampled from an LLM tends to occupy negative curvature regions of the model's log probability function. Leveraging this observation, we then define a new curvature-based criterion for judging if a passage is generated from a given LLM. This approach, which we call DetectGPT, **does not require training a separate classifier, collecting a dataset of real or generated passages, or explicitly watermarking generated text**. It uses only log probabilities computed by the model of interest and random perturbations of the passage from another generic pre-trained language model (e.g., T5). We find DetectGPT is more discriminative than existing zero-shot methods for model sample detection, notably improving **detection of fake news articles** generated by 20B parameter GPT-NeoX from 0.81 AUROC for the strongest zero-shot baseline to 0.95 AUROC for DetectGPT. See ericmitchell.ai/detectgpt for code, data, and other project information.

The Startup Idea

You just completed a research thesis on detecting machine-generated text using probability curvature — no training data, no watermark, no classifier.

You showed 0.95 AUROC distinguishing human news articles from ones generated by a 20B open-weights model, beating the best prior zero-shot method (0.81) on several benchmarks.

Two friends are excited about the application to academic integrity.

Idea: Let's commercialize it and sell it to universities and publishers.



Pangram.



Credits



11 left

Unlock Free Trial

Share



Text



Image

HISTORY



All Content



More



Pangram LMS



View Page Current

2xp1.pdf

🕒 August 24 2026 · 888 words



AI Highlight



Mixed

These studies also mostly estimate a point-in-time effect and say little about how a rollout unfolds within an organization—what a headline metric measures, what drives it, and how the rollout reshapes the organization. Qualitative work supplies some mechanisms, such as the “productivity pressure paradox” by which a mandate imposed without enablement undermines its own goal [11], but does not follow outcomes across a rollout; joining the two is scarce because it requires a documented mandate and multi-year telemetry from an identifiable firm. Abstract—Enterprises increasingly mandate AI coding tools and report large productivity gains, yet longitudinal evidence on how such a mandate unfolds is scarce. In this paper, we present a quantitative case study of a documented enterprise “2→” mandate at a mid-sized, AI-forward company that has been committed to doubling merged pull requests per engineer since mid-2025. In a panel of 802 developers and 196,212 pull requests (January 2024–April 2026), per-capita throughput eventually doubled, reaching



Overview



Details



Notes



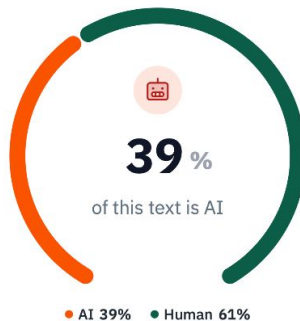
AI Detected



Pangram 4.0

888 words scanned

We believe that this text is a mix of AI and human-written content.



AI-generated content appears throughout



Was this result helpful?



Plagiarism Analysis

Protect authenticity in every industry

Pangram provides AI detection solutions for educators, publishers, law firms, HR departments, foundations, and more.

[Contact Us](#)

AI Detector for Education



For teachers

Pangram's transparency tool reliably detects AI content and checks for plagiarism to help teachers understand the origin of written text and maintain integrity and fairness in assessment.

AI Detector for Enterprise



For higher ed

Pangram's LMS integrations allow universities to create informed AI policies. Our AI detector's tool accuracy gives them the peace of mind to implement those policies at scale.

Protect authenticity in every industry

Pangram provides AI detection solutions for educators, publishers, law firms, HR departments, foundations, and more.

Contact Us



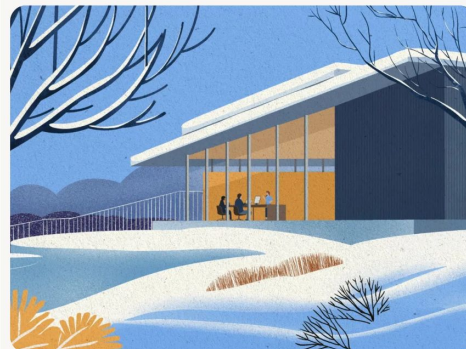
AI Detector for Education

AI Detector for Enterprise



For publishers

AI-generated text can be used to churn out low-quality articles at an unprecedented rate, overwhelming publishers with submissions and making it difficult to identify high-quality, original content. AI-generated images could violate copyright or produce content that's inappropriate or off-brand. Pangram helps publishing companies ensure the originality and authenticity of content at scale.



For content moderation

AI enables bad actors to flood platforms with fake reviews, misleading product descriptions, and fraudulent listings, negatively affecting both consumers and legitimate sellers. It can also be used to fabricate medical advice, legal guidance, and financial tips, putting users at serious risk. Pangram helps companies ensure safety and quality control of user generated content at scale.

For law firms

Gen AI introduces regulatory, ethical, and professional responsibility and liability risks to legal practice. Pangram helps law firms ensure that Gen AI isn't being used by their lawyers and staff in unauthorized or undisclosed ways.

For HR

AI makes it easy to fake application materials and generate false work samples. Pangram helps HR departments ensure job applicants are being honest with their application materials.

For foundations and philanthropies

Many foundations have developed policies restricting Gen AI use in grant applications. Pangram helps foundations enforce those policies at scale, so they can focus their limited resources on applicants who share their values.

Breakout: Likely challenges in building commercial product?

As a group, think about challenges that the team will likely face when turning their research into a product:

- One machine-learning challenge
- One engineering challenge in building the product
- One challenge from operating and updating the product
- One team or management challenge
- One business challenge
- One safety or ethics challenge

*Post answer to #lecture on Slack and tag all group members **and @mlipbot** (skip those who have not set up slack yet). Follow up to @mlipbot comment.*



Guidance on AI Detection and Why We're Disabling Turnitin's AI Detector

Aug 16, 2023, 7:00 AM

Share



In April of this year, Turnitin released an update to their product that reviewed submitted papers and presented their determination of how much of a paper was written by AI. [As we outlined at that time](#), many people had important concerns and questions about this new tool, namely how the product exactly works and how reliable the results would be. After several months of using and testing this tool, meeting with Turnitin and other AI leaders, and talking to other universities who also have access, ***Vanderbilt has decided to disable Turnitin's AI detection tool for the foreseeable future.*** This decision was not made lightly and was made in pursuit of the best interests of our students and faculty.



Opinion

GPT detectors are biased against non-native English writers

Weixin Liang,^{1,4} Mert Yuksekgonul,^{1,4} Yining Mao,^{2,4} Eric Wu,^{2,4} and James Zou^{1,2,3,*}

¹Department of Computer Science, Stanford University, Stanford, CA, USA

²Department of Electrical Engineering, Stanford University, Stanford, CA, USA

³Department of Biomedical Data Science, Stanford University, Stanford, CA, USA

⁴These authors contributed equally

*Correspondence: jamesz@stanford.edu

<https://doi.org/10.1016/j.patter.2023.100779>

GPT detectors frequently misclassify non-native English writing as AI generated, raising concerns about fairness and robustness. Addressing the biases in these detectors is crucial to prevent the marginalization of non-native English speakers in evaluative and educational settings and to create a more equitable digital landscape.

Introduction

Generative language models based on GPT, such as ChatGPT, have gained significant attention in recent times. Within a mere 2 months of its launch, ChatGPT amassed over 100 million monthly active users, marking its place as one of the fastest-growing consumer internet applications in history.¹ Despite their immense potential for enhancing productivity and

in GPT detectors against non-native English speakers: over half of the non-native English writing samples were misclassified as AI generated, while the accuracy for native samples remained near perfect. Our analysis further revealed a trend where more literary language was classified as more “human”: enhancement of word choice in non-native English writing samples reduced misclassifica-

demonstrated by their high misclassification of TOEFL essays written by non-native speakers. In our study, we evaluated the performance of seven widely used GPT detectors on 91 TOEFL (Test of English as a Foreign Language) essays from a Chinese forum and 88 US eighth-grade essays from the Hewlett Foundation’s ASAP dataset. While the detectors accurately classified the US student

Examples for discussion

- What does "accuracy" even mean here?
- What is the ground truth? Who labels it, and can that label ever be verified?
- Is a student who used Grammarly cheating? Who asked for an outline? Who wrote a draft and asked for edits?
- Which error is worse — a missed cheater or a falsely accused student? By how much? Who decides the threshold, us or the university?
- What accuracy do our customers think they're buying when we say 99%?
- How do we present a probability to a professor who wants a yes/no?
- When a student disputes a flag, what do we tell them? Can we explain the decision?
- Do we ever see whether we were right? What feedback signal do we actually get?

Examples for discussion 2

- A new frontier model ships on Monday. Our detector has never seen it. What happens Tuesday?
- How fast can we retrain and redeploy? What's our latency from "model released" to "we detect it"?
- A "humanizer" service launches specifically to beat us. Now what?
- Every LMS wants an integration (e.g., Canvas). Who maintains those?
- Peak load = every Sunday night at 11pm, every semester's end. Can we handle it? How to handle large amounts of data per day? Buy more machines or move to the cloud?
- We improve overall accuracy but false positives rise on one subpopulation. Did we get better?
- How can we see that everything is going fine and page somebody if it is not?
- Tensorflow update; does our infrastructure still work?
- Can we offer AI detection as an app? As a web service?
- Can we get better the longer a user interacts with us? Will this benefit the next audit they run as well?

Examples for discussion 3

- How many languages? How many domains — essays, code, résumés, product reviews, legal filings, peer reviews?
- Short text? A 40-word discussion post? A tweet?
- Can we tell which model wrote it? Should we claim that?
- Can we learn from customer submissions? Are we allowed to keep student essays?
- How do we debug a false positive on an essay we're contractually forbidden from retaining?
- Can a competitor probe our API to build an evasion tool? Can they just buy a subscription and do it?
- If we publish our benchmark, do we hand adversaries a target?

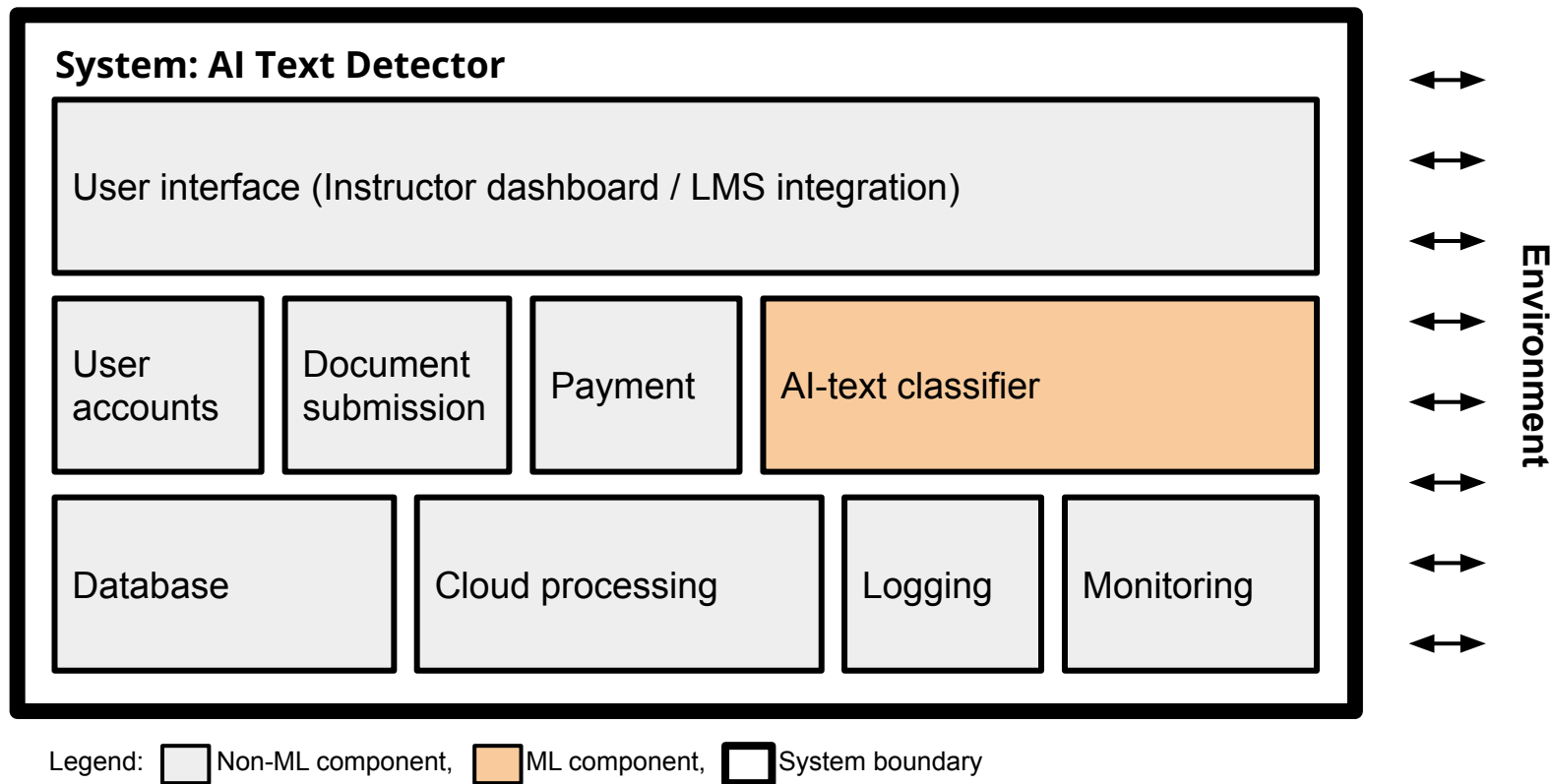
What qualities are important for a good commercial AI-text detector?



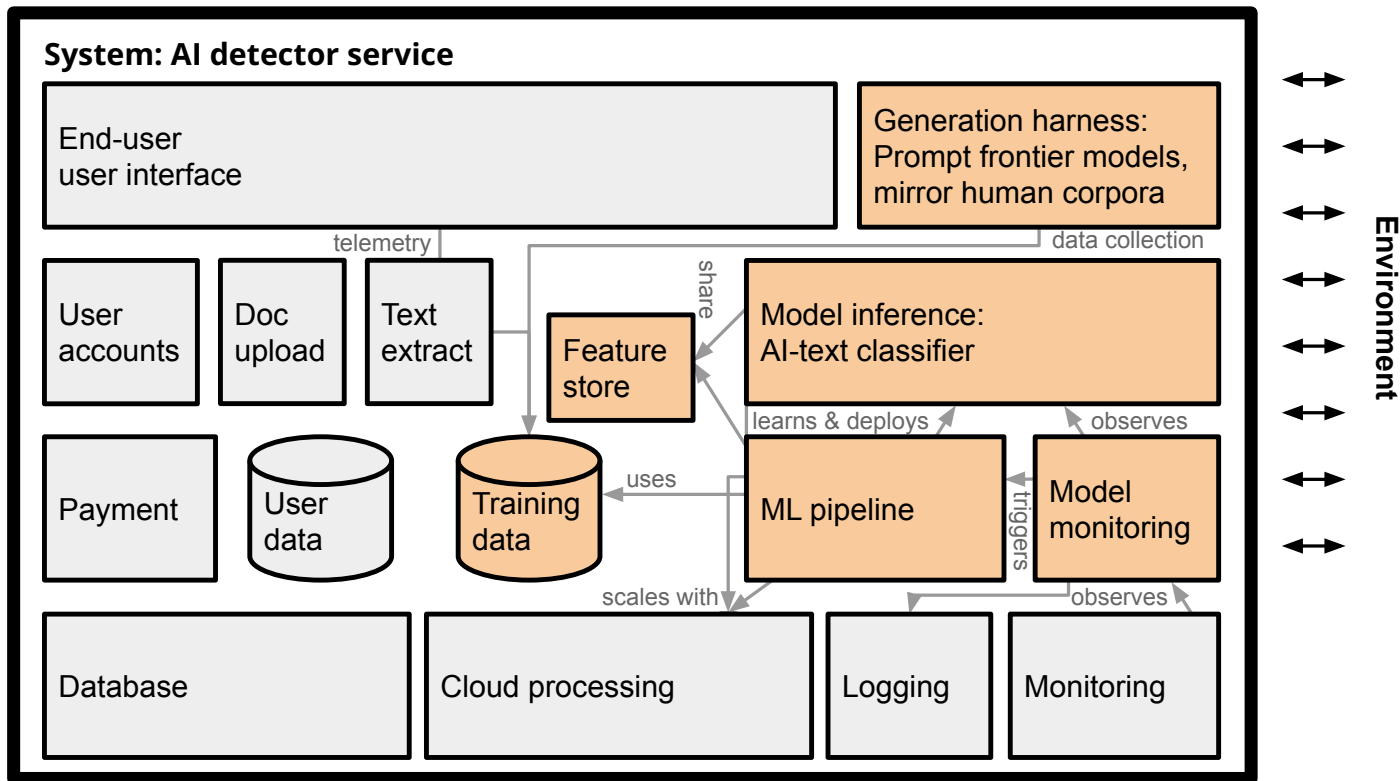
What non-ML components are needed?



ML in a Production System

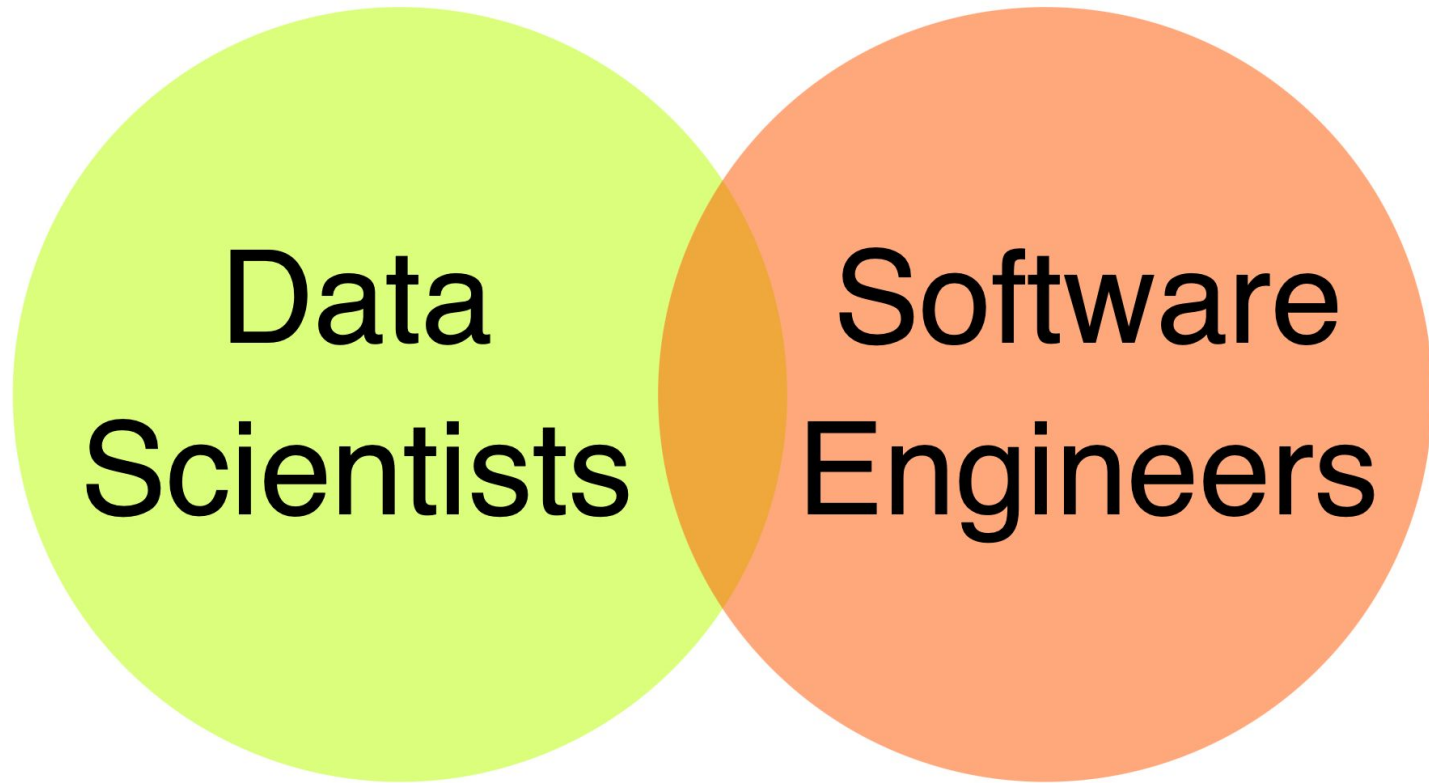


ML in a Production System



Legend: Non-ML component, ML component, system boundary

Development Teams



and Data engineers + Domain specialists + Operators + Business team + Project managers + Designers, UI Experts + Safety, security specialists + Lawyers + Social Scientists + ...

Data scientist

- Often fixed dataset for training and evaluation (e.g., XSum news articles + GPT-NeoX generations)
- Focused on accuracy
- Prototyping, often Jupyter notebooks or similar
- Expert in modeling techniques and feature engineering
- Model size, updateability, implementation stability typically does not matter

Software engineer

- Builds a product
- Concerned about cost, performance, stability, release time
- Identify quality through customer satisfaction
- Must scale solution, handle large amounts of data
- Detect and handle mistakes, preferably automatically
- Maintain, evolve, and extend the product over long periods
- Consider requirements for security, safety, fairness

Likely collaboration challenges?



What might Software Engineers and Data Scientists focus on?

The screenshot displays the Pangram application interface. At the top, the Pangram logo and a credit bar (11 left, Unlock Free Trial) are visible. The left sidebar contains navigation options: Text, Image, HISTORY, and All CI. The main document area shows '2xp1.pdf' with a 'Mixed' classification and a date of August 24, 2026, for 888 words. An 'AI Highlight' button is present. The document text discusses productivity pressure paradox and AI coding tools. On the right, an 'AI Detected' panel shows a 39% AI detection rate for 888 words scanned, with a donut chart and a legend for AI (39%) and Human (61%). Below the chart, a message states 'AI-generated content appears throughout' with a right arrow. At the bottom of the panel, a feedback prompt asks 'Was this result helpful?' with thumbs up/down icons and a close button.

Pangram.

Credits 11 left Unlock Free Trial Share

2xp1.pdf Mixed

August 24 2026 · 888 words

AI Highlight

These studies also mostly estimate a point-in-time effect and say little about how a rollout unfolds within an organization—what a headline metric measures, what drives it, and how the rollout reshapes the organization. Qualitative work supplies some mechanisms, such as the “productivity pressure paradox” by which a mandate imposed without enablement undermines its own goal [11], but does not follow outcomes across a rollout; joining the two is scarce because it requires a documented mandate and multi-year telemetry from an identifiable firm.

Abstract—Enterprises increasingly mandate AI coding tools and report large productivity gains, yet longitudinal evidence on how such a mandate unfolds is scarce. In this paper, we present a quantitative case study of a documented enterprise “2→” mandate at a mid-sized, AI-forward company that has been committed to doubling merged pull requests per engineer since mid-2025. In a

Overview Details Notes

AI Detected Pangram 4.0

888 words scanned

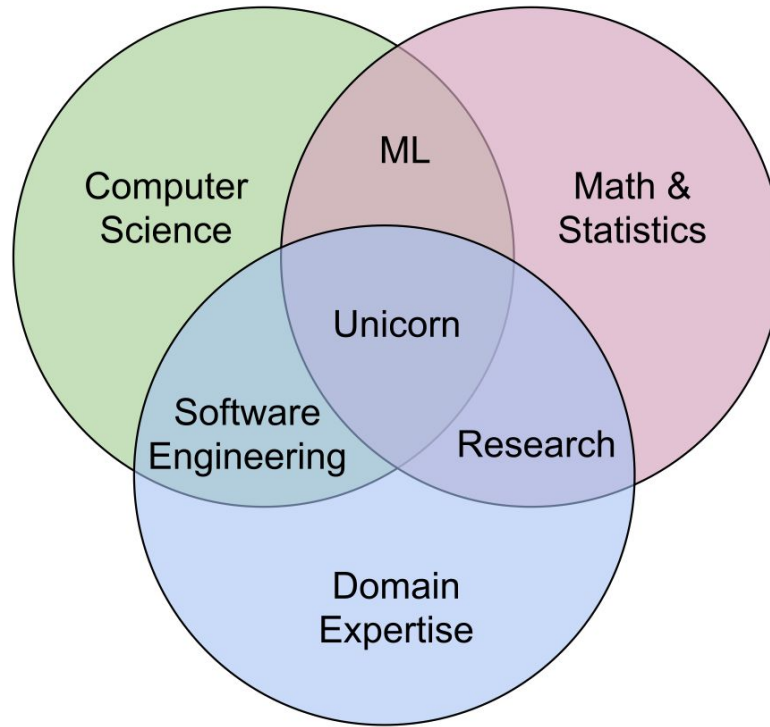
We believe that this text is a mix of AI and human-written content.

39% of this text is AI

AI 39% Human 61%

AI-generated content appears throughout

Was this result helpful? 👍 👎 ✕



By Steven Geringer, via Ryan Orban. Bridging the Gap Between Data Science & Engineer: Building High-Performance Teams. 2016

T-Shaped People

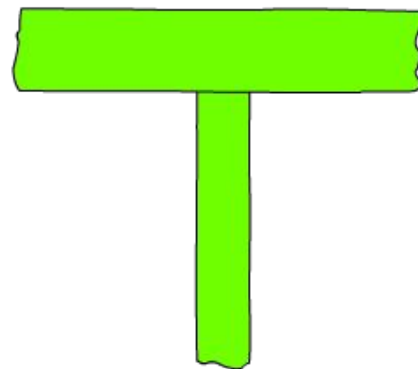
Broad-range generalist + Deep expertise



"I-shaped"
Expert at one thing



Generalist
Capable in a lot of things
but not expert in any



"T-shaped"
Capable in a lot of things
and expert in one of them

Figure: Jason Yip. Why T-shaped people?. 2018

T-Shaped People

Broad-range generalist + Deep expertise

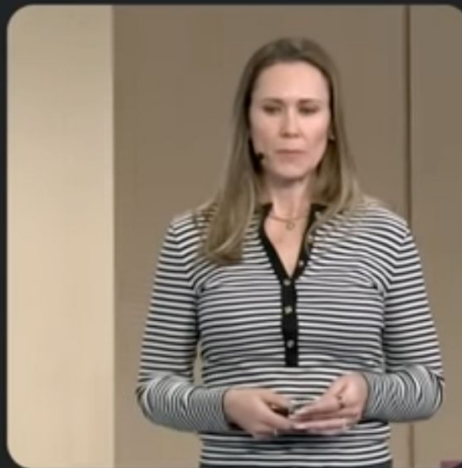
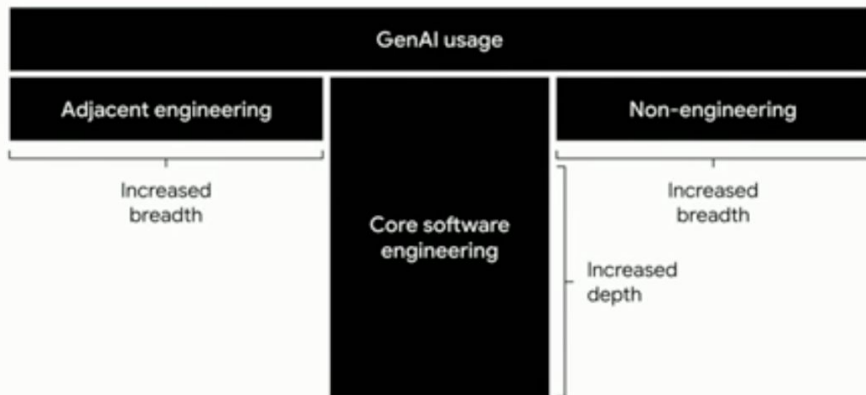
Example:

- Basic skills of software engineering, business, distributed computing, and communication
- Deep skills in deep neural networks (technique) and medical systems (domain)

Latest Buzzword: π -Shaped People



The T-shaped developer



Macvean & Forsgren. Build core skills to thrive as an AI-era developer. Google I/O 2026



Syllabus and Logistics

Joining late

We have cleared most waitlists.

Please email the course admin Jenni Cooper (cooperj@andrew.cmu.edu), not us, to inquire about WL status/registration movement in.

If and when you join late:

1. Post an introduction on Slack (#social)
2. See the Canvas front page about automatic extensions

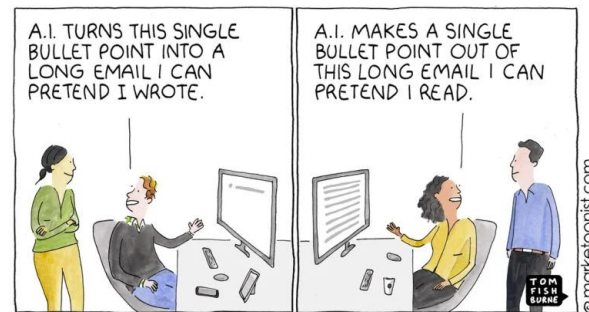
Syllabus and Class Structure

17-445/17-645/17-745, Fall 2026, 12 units

Tuesday/Thursday 3:30-4:50pm

Recitation Fridays 9:30am, 11am, 12:30pm, 2pm, 3:30pm
(and Thursday 6pm, online)

Communication



Preferably ping us on Slack (link on Canvas)

Email also okay, but ask your chatbot to keep the email short

All announcements through Slack #announcements

Weekly office hours, starting Thursday, schedule on Canvas

Post questions on Slack – Please use #general or #assignments and post publicly if possible; your classmates will benefit from your Q&A!

All course materials (slides, recordings, assignments, old midterms) available on course website: <https://mlip-cmu.github.io/f2026/>

Office Hours

Times posted soon on Canvas

Feel free to drop in with questions or just to chat

Some office hours have 15 min signup slots

Class with engineering flavor

Focusing on engineering judgment

Arguments, tradeoffs, and justification, rather than single correct answer

Practical engagement, building systems, testing, automation

Strong teamwork component

Assignments focused on building and operating things, not math



Fundamentals of Engineering AI-Powered Software Systems

Holistic system view: AI and non-AI components (models, LLMs, agents), pipelines, stakeholders, feedback loops

Requirements:

System and model goals
User requirements
Environment assumptions
Quality beyond accuracy
Measurement
Risk and hazard analysis
Planning for mistakes

Architecture + design:

Modeling tradeoffs
Deployment architecture
Data science pipelines
Telemetry, monitoring
Anticipating evolution
Big data processing
Human-AI design

Quality assurance:

Model & agent testing
Data quality
QA automation
Testing in production
Infrastructure quality
Debugging

Operations:

Continuous deployment
Contin. experimentation
Configuration mgmt.
Monitoring
Versioning
Big data
DevOps, MLOps

Teams and process: Data science vs software eng. workflows, interdisciplinary teams, collaboration points, technical debt

Responsible AI Engineering

Provenance,
versioning,
reproducibility

Safety

Security and
privacy

Fairness

Interpretability
and explainability

Transparency
and trust

Risk, ethics, governance, regulation, compliance, organizational culture

Thoughts on Generative AI for Homework?



ChatGPT, Copilot, Claude Code, ...? Homework, labs, midterms, ...?

Our Position on Generative AI for Homework

This is a course on responsible building of ML products. This includes questions of how to build generative AI tools responsibly and discussing what use is ethical.

Feel free to use them and explore whether they are useful.

We assume **(and recommend)** you are using AI coding tools for assignments

Warning: Be aware of hallucinations and AI slop. Requires understanding to check answers. We test them; they often need extra hand-holding or generate **bad/wrong answers**. (Last semester 80% of students lost 20+% points on individual assignment due to not critically checking AI output of I3)

You are responsible for the correctness of what you submit!

Our Position on Generative AI for Homework II

Even if you generate correct solutions, we expect you to understand them

Every assignment has a requirement to explain your solution to the course staff and to answer questions probing your understanding
(You can attempt this multiple times if needed)

Two strict **restrictions on AI** use:

1. exams and any on-paper quizzes we designate as strictly offline
2. live conversations with course staff, whether over Zoom or in person

AI Tooling

We do not recommend specific AI models or tools
AI for coding is beyond the scope of what we teach

Consider using commercial state of the art agentic tools (e.g. Claude Code) and paying \$20-100/month for a subscription (roughly comparable to textbook costs)

Prerequisites

Some machine learning experience required

- Basic understanding of data science process, incl. data cleaning, feature engineering, using ML libraries
- High level understanding of machine-learning approaches
 - Supervised learning
 - Regression, decision trees, neural networks
 - Accuracy, recall, precision, ROC curve
- Ideally, some experience with notebooks, sklearn, or other frameworks

Basic programming and command-line skills will be needed

No further software-engineering knowledge required

- Teamwork experience in product team is useful but not required
- No required exposure to requirements, software testing, software design, continuous integration, containers, process management, etc.
 - If you are familiar with these, there will be some redundancy – sorry!

First Homework Assignment I1

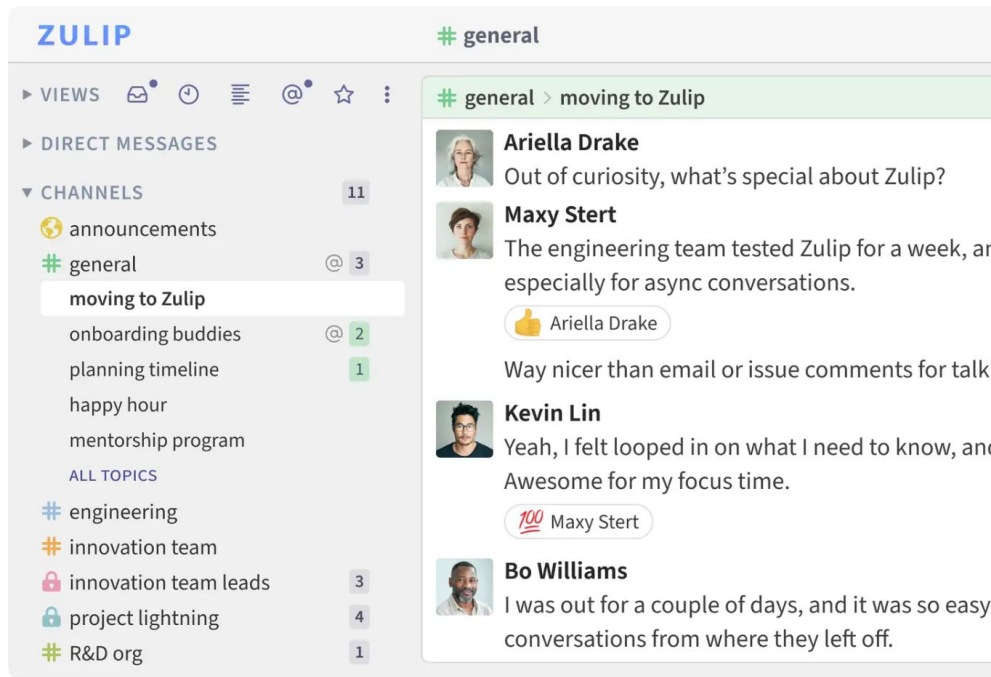
“Coding warmup assignment”

Out now, due Tuesday Sep 1

Enhances Slack-style mature web application with two LLM-powered features

Coding assignment in nontrivial code base, front-end and backend, change existing code, learn new APIs, handle credentials

Use of AI coding tools encouraged



Active lecture

Case study driven

Discussions highly encouraged

Regular in-class activities, breakouts

Contribute your own experience!

Discussions over definitions

The screenshot displays the Pangram AI detection interface. On the left is a sidebar with navigation options: Text, Image, HISTORY, and All C. The main content area shows a document titled '2xp1.pdf' with a timestamp of August 24, 2026, and 888 words. A 'Mixed' label is present. The document text is displayed with an 'AI Highlight' feature. The text reads: 'These studies also mostly estimate a point-in-time effect and say little about how a rollout unfolds within an organization—what a headline metric measures, what drives it, and how the rollout reshapes the organization. Qualitative work supplies some mechanisms, such as the “productivity pressure paradox” by which a mandate imposed without enablement undermines its own goal [11], but does not follow outcomes across a roll-out; joining the two is scarce because it requires a documented mandate and multi-year telemetry from an identifiable firm. Abstract—Enterprises increasingly mandate AI coding tools and report large productivity gains, yet longitudinal evidence on how such a mandate unfolds is scarce. In this paper, we present a quantitative case study of a documented enterprise “2→” mandate at a mid-sized, AI-forward company that has been committed to doubling merged pull requests per engineer since mid-2025. In a panel of 802 developers and 196,212 pull requests (January 2024–April 2026), per-capita throughput eventually doubled, reaching

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Grading

- 35% individual assignment (5 one-week assignments, 100pt each)
- 30% group project with final presentation (4 milestones, 415pt total)
- 25% two in-class exams (or 15% if opting in to graded participation)
- 0% participation (or 10% if opting in to graded participation)
- 10% labs
- No final exam (final presentations will take place in that time slot)

Grade cutoffs in syllabus (>82% B, >94 A-, >96% A, >99% A+)

Recordings and Attendance

Best effort lecture recordings, textbook, and slides available

Strongly recommend to attend lecture – discussions are important to learning

[insert warning about relying on recordings]

Participation can be part of your grade (opt-in within 4 weeks, by Sep 22 – use `/participation optin` in Slack)

Contact us for accommodations (illness, interview travel, unforeseen events), or have your advisor reach out. We try to be flexible.

Participation != Attendance

Expecting active participation at minimum in breakout discussions (1-2 per lecture)

- Group activity, in person! Solo and remote participation does not count
- Post answer to Slack, tag group members and @mlipbot and follow up on feedback

When not feeling well—please stay home and get well, and email us for accommodation

Pretending to participate, tagging others who do not participate is an academic integrity violation and will result in a negative participation grade for all involved. Do not opt in if you do not plan to regularly participate.

Grading (if opted in)

- 100%: Participates actively at least once in most lectures (4 lectures waived, no questions asked)
- 80%: Participates actively at least once in two thirds of the lectures
- 60%: Participates actively at least once in over half of the lectures
- 30%: Participates actively at least once in one quarter of the lectures
- 0%: Participation in less than one quarter of the lectures.

Important dates

Tuesday, Sep 15, 3:30-4:50: Team announcement and first meeting

Thursday Oct 8, 3:30-4:50: Midterm 1

Thursday Dec 4, 3:30-4:50: Midterm 2

pen&paper midterms during lecture time, not cumulative

find old midterms on the course website

Final presentations in final exam slot (tbd)

may be as late as Dec 13; do not book flights before dates
announced at <https://www.cmu.edu/hub/docs/final-exams.pdf>

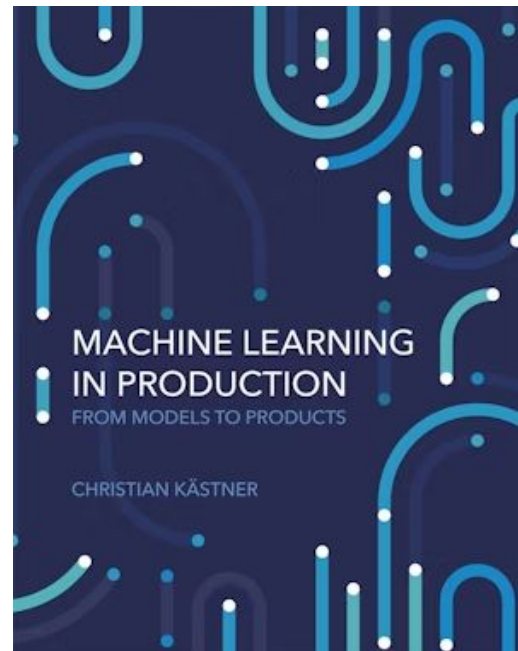
Readings

We will selectively assign readings (papers, blog posts, podcasts), about 1 per week

You will discuss/apply readings during the lecture

We might ask about readings content in the midterm

Optional: Supplementary textbook, closely tracks lecture content. Published online at <https://mlip-cmu.github.io/book/>, printed through MIT Press



Assignments

Series of 5 small(ish) **individual assignments**

- Build LLM-powered features for chat application (engineering warmup)
- Build a movie recommendation model and address the cold-start problem with an LLM (ML warmup)
- Create a LLM-powered hazard analysis tool and apply it for a smart dashcam system
- Secure an AI agent for customer service support
- Create end-user explanations for a medical device

Large **team project** with 4 milestones:

- Build and deploy a movie recommendation service
- Testing in production, monitoring, incidence response
- Final presentation

Deadlines always Tuesday night; see schedule

Check-ins and debriefing

After every individual assignment, mandatory (graded) check-in with TA

- Within 2 weeks of submission
- Explain your solution, answer questions, discuss reflection questions
- Graded pass/fail on the spot, can retry
- During office hours, we will schedule extra OH and signup times

After every group project milestone, mandatory (graded) debriefing with TA

- Discuss solution and teamwork

17-745 PhD Research Project

Research project instead of individual assignments I4 and I5

Design your own research project and write a report

- A case study, empirical study, etc.

Very open ended: Align with own research interests and existing projects; requires preliminary results and final presentation

See the [project requirements](#) and deadlines and talk to us

Labs

Please attend the lab for which you are registered.

Introducing various tools, e.g., Kafka (stream processing), Jenkins (continuous integration), MLflow (experiment tracking), Docker & Kubernetes (containers), Prometheus & Grafana (monitoring), SHAP (explainability)...

Hands on exercises, bring a laptop

Introducing tools useful for assignments

About 1h of work, graded pass/fail, low stakes, show & explain work to TA

First lab **this Friday**: Kafka stream processing for I2

Lab grading and collaboration

We recommend that you start each lab before the lab session, but it can be completed during

Graded pass/fail by TA on the spot, can retry

Relaxed collaboration policy: Can work with others before and during recitation, but have to show & explain solution to TA individually

(Think of labs as mandatory office hours)

Lab online options (experimental)

If you complete the lab ahead of time, you may be able to show it over Zoom on Thursday evenings

You will show/explain your work to the TA over Zoom

TA will grade, but not provide help; if you like help or fail grading, attend your lab session in person on Friday

Logistics on how and when to queue will be posted on Slack

Grading Philosophy

Specification grading, based in adult learning theory

We are making every effort to be clear about expectations (specifications), will clarify if you have questions

Assignments broken down into expectations with point values, each graded **pass/fail**

Opportunities to resubmit work until last day of class

[\[Example\]](#)

Token System for Flexibility

8 individual tokens per student:

- Submit individual assignment 1 day late for 1 token (after running out of tokens, 15% penalty per late day)
- Redo individual assignment for 3 token and a 10% tax
- Late/redo lab, team feedback, ... for 1 token
- Extra participation day for 1 token

8 team tokens per team:

- Submit milestone 1 day late for 1 token
- Redo milestone for 3 token and a 10% tax

How to use tokens

No need to tell us if you plan to submit very late. We will assign 0 and you can resubmit

Resubmit through Canvas – you will get a confirmation comment

Request token use for late lab during office hours

#remaining tokens reflected on Canvas

Accommodations

The token system provides built-in flexibility and will cover many issues

Slack/email for accommodations, like extensions or waived participation

- Usually exceptional, unanticipated circumstances
- Typical: medical issues, mental health issues, interviews, emergencies
- We usually ask you to also inform your advisor

Group project

Instructor-assigned teams; assignments announced during Feb 2 lecture

Teams stay together for project throughout semester.

There will be a Catme Team survey around Jan 28 (3pt)

Some advice in lectures; we'll help with debugging team issues

TA assigned to each team as mentor; mandatory debriefing with mentor and peer grading on all milestones (based on citizenship on team)

Bonus points for social interaction in project teams

Academic honesty

- See web page; in a nutshell: do not copy from other students, do not lie, do not share or publicly release your solutions
- In group work, be honest about contributions of team members, do not cover for others
- *Do not falsely pretend to participate in in-class participation (in person, in group)*
- *Do not use AI for exams & in-person/zoom check-ins with TA*
- Collaboration okay on labs, but not for individual assignments or exams
- If you feel overwhelmed or stressed, please come and talk to use (see syllabus for other support opportunities)

Education with AI Agents?

The Problem

Homework is a sweet spot for AI

AI restrictions hard to enforce even if we tried

The token system gives flexibility to fix AI mistakes

Easy to avoid any learning

The Point of Homework

Not “The world needs more essays,” “we needed one more quicksort implementations,” or “we really need a solution to this math problem”
– the deliverable is not the learning

Building component skills that provide the foundation for
higher-level skills, build judgment and taste

Junior to senior engineer pipeline

What’s your VARP (“Value over replacement player”) or VARAI?



escu • Cam



Value of a senior engineer

Bigger picture, system judgement, taste

Oversight, detecting problems early, responsible judgment

Judgement about where direction and oversight is needed

Managing uncertainty and risk

– usually gained through experience over time

“Every time we hand work to a wizard, we lose a chance to develop our own expertise, to build the very judgment we need to evaluate the wizard’s work.

We’re getting something magical, but we’re also becoming the audience rather than the magician, or even the magician’s assistant.”

Mollick, E. On working with wizards. One Useful Thing (Sept. 10, 2025); <https://www.oneusefulthing.org/p/on-working-with-wizards>

Our view

You can probably prompt yourself to an A with little thinking in this course

We provide the opportunities and safety net for actual learning

Experiment and calibrate trust with AI use

Do not make yourself replaceable

What makes software with ML challenging?

Lack of Specifications

```
/**  
    Return true if the text was written by AI  
    ????  
*/  
boolean isAIGenerated(String text);
```

Traditional SE: specify what to do & how to test

ML usually black boxes (maybe justified, for their complexity...)

Even if you put specs in, e.g., LLM prompts, unclear if they will follow

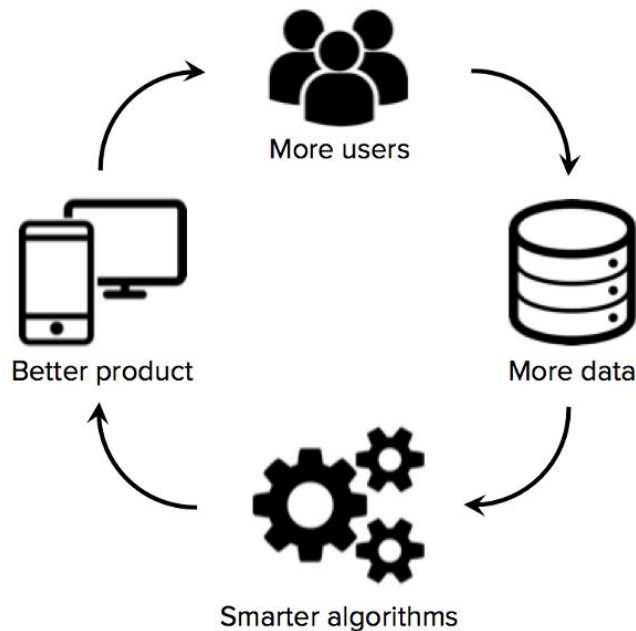
Agents combine multiple unreliable components

Data Focused and Scalable

MLs get the “specs” from data; Larger the better

Deductive reasoning (applying logic rules) to
Inductive Reasoning (generalizing from observation)

Cause scalability issues



ML Models Make Mistakes

...often in unexpected ways

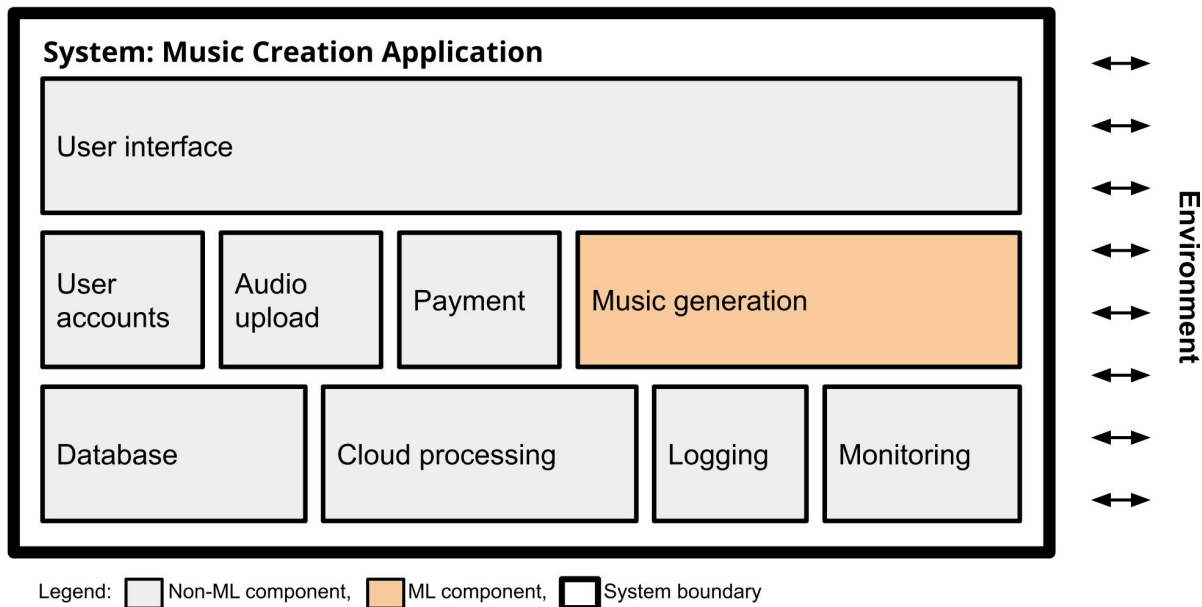
Hard to foresee and capture because no spec

What does it mean to be correct? Can only evaluate whether it works well enough (on average) on some test data!



NeuralTalk2: A flock of birds flying in the air
Microsoft Azure: A group of giraffe standing next to a tree
Image: Fred Dunn, <https://www.flickr.com/photos/gratapictures> - CC-BY-NC

Interaction with the environment



Our system must be able to tolerate some incorrect predictions, and be aware of how it might influence the world...

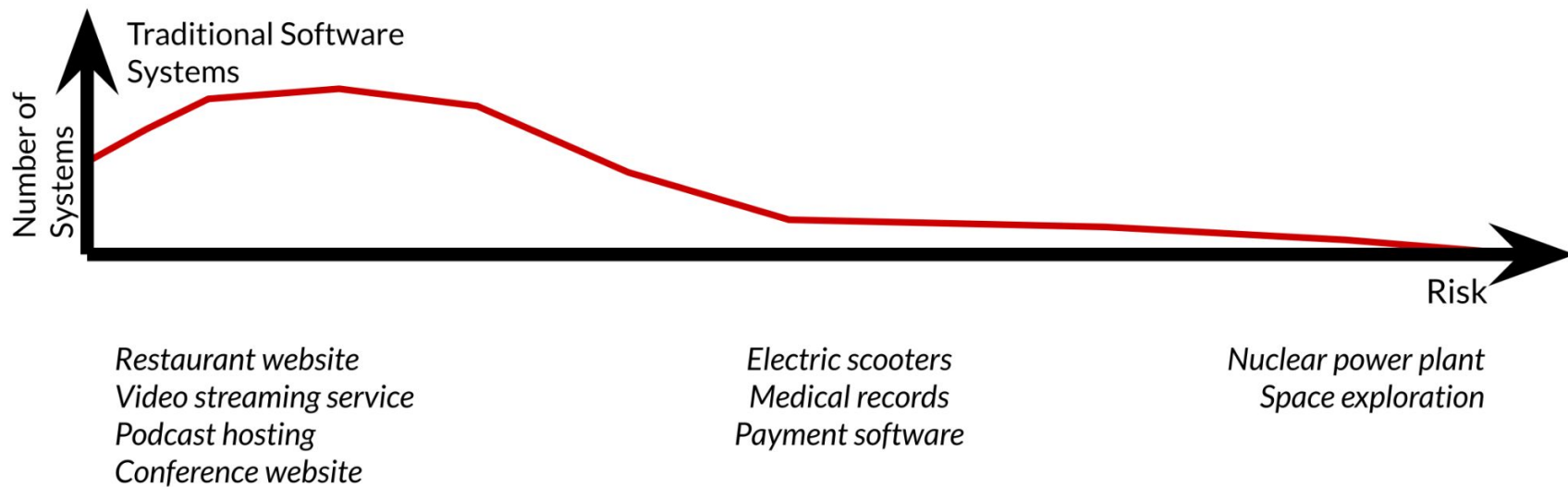
It's not all new

We routinely build:

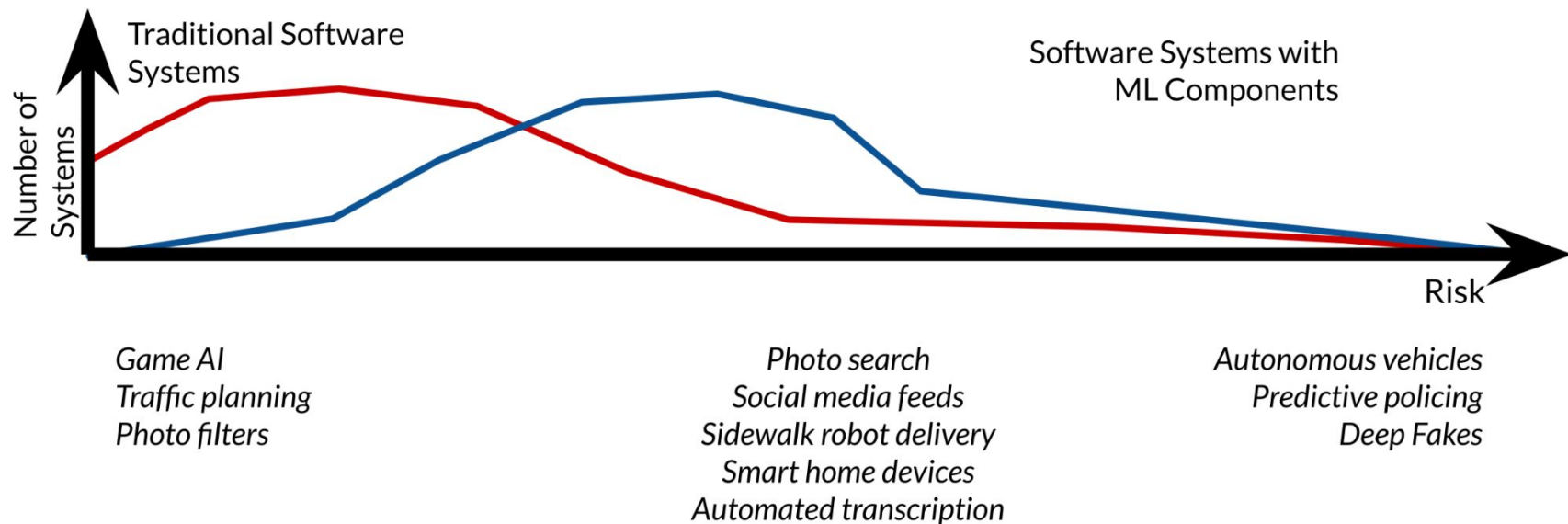
- Safe software with unreliable components
- Non-ML big data systems, cloud systems
- “Good enough” and “fit for purpose” not “correct”
- Cyberphysical systems

ML intensifies our challenges

Complexity



Complexity



Introductions

Before the next lecture, introduce yourself in Slack channel #social:

- Your (preferred) name
- In 1-2 sentences, describe your data science background and goals (e.g., coursework, internships, work experience)
- In 1-2 sentences, describe your software engineering background, if any, and goals (e.g., coursework, internships, work experience)
- The kind of job you are aiming for after graduation
- A hobby or a favorite activity outside school

Summary

Machine learning components are part of larger systems

Data scientists and software engineers have different goals and focuses

- Building systems requires both
- Various qualities are relevant, beyond just accuracy

Machine learning brings new challenges and intensifies old ones